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Connectivity in graphs;

Theorem 1; a graph is bipartite $\Leftrightarrow$ every cycle of G has even length.

proof;

$\Rightarrow$ since G is bipartite, we can split its vertex set into two disjoint sets A and B so that each edge of G joins a vertex of A and a vertex of B . let $v_0 \rightarrow v_1 \rightarrow \dots \rightarrow v_n \rightarrow v_0$ be a cycle of G and assume v_0 in A then v_1 in B , since v_n must be in B then, the cycle has even length.

$\Leftarrow$: pick any vertex $u \in V(G)$, define two sets

$X = \{x \in V; \text{distance from } u \text{ to } x \text{ is even}\}; Y = \{y \in V; \text{distance from } u \text{ to } y \text{ is odd}\}.$

distance $d(u, x)$: length of shortest path from u to x .

(X, Y) is bipartition, no edge connects two vertices in X or two in Y .

suppose for contradiction: there exists an edge connecting two vertices in the same set, say $v, w \in X$.

let P = shortest path from u to v , Q : shortest path from u to w .

denote u_f : last common vertex of P and Q (where paths diverge).

$P = u \rightarrow \dots \rightarrow u_f \rightarrow p_1 \rightarrow v$; $Q = u \rightarrow \dots \rightarrow u_f \rightarrow q_1 \rightarrow w$

both P and Q has the same length toward u_f , since length of P and Q is even, then length of P_1 and Q_1 is even (because total is even). now consider the path $P_1 + \text{reverse of } Q_1 + \text{edge } (v, w)$ this is a cycle of odd length (contradiction) (proceed the same for set Y).



1. Theorem 2: let G be a simple graph on n vertices. if G has k components, then the number m of edges of G satisfies:

$$n - k \leq m \leq \frac{(n - k)(n - k + 1)}{2}$$

proof: we prove the lower bound $m \geq n - k$ by induction on the number of edges, the result is trivial if G is a null graph.

if G contain a few edges as possible (say m_0), then the removal of any edge of G must increase the number of components by 1, the remaining graph contain n vertices, $k+1$ components, $m-1$ edges. by hypothesis: $m_0 - 1 \geq n - (k+1) \Rightarrow m_0 \geq n - k$ as required.

to prove the upper bound, we assume that each component of G is complete, suppose that there are two components C_i, C_j with n_i and n_j vertices respectively, $n_i \geq n_j \geq 1$, if we replace C_i and C_j by complete graphs on n_i+1 and n_j-1 vertices, then the total number of edges $\frac{(n_i+1)(n_i)}{2} + \frac{(n_j-1)(n_j-2)}{2}$ is increased by $n_i - n_j + 1 > 0$. it follows that, in order to maximize number of edges, G must contain a complete graph on $n - k + 1$ vertices and $k - 1$ isolated vertices. which gives: $\frac{(n - k + 1)(n - k)}{2} + 0 \times (k - 1)$.

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Corollary 3: Any simple graph with n vertices and more than $\frac{(n-1)(n-2)}{2}$ edges is connected.

proof:

$$\frac{(n-1)(n-2)}{2} \geq \frac{(n-k)(n-k+1)}{2} ; \forall k \geq 2$$

(by Theorem 2).

$$\Rightarrow k = 1.$$